Contest Duration: 2018-01-14(Sun) 10:00 - 2018-01-14(Sun) 12:10 (local time) (130 minutes)

C - Median Sum

Editorial

Time Limit: 2 sec / Memory Limit: 512 MiB

Score : 700 points

Problem Statement

You are given N integers $A_1, A_2, ..., A_N$.

Consider the sums of all non-empty subsequences of A. There are $2^N - 1$ such sums, an odd number.

Let the list of these sums in non-decreasing order be $S_1, S_2, ..., S_{2^N-1}$.

Find the median of this list, $S_{2^{N-1}}$.

Constraints

- $1 \leq N \leq 2000$
- $1 \leq A_i \leq 2000$
- All input values are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
A_1 \quad A_2 \quad ... \quad A_N
\end{array}
$$

Output

Print the median of the sorted list of the sums of all non-empty subsequences of A.

Sample Input 1

Copy

```
3
1 2 1
```

Sample Output 1

Copy

```
2
```

In this case, $S = (1, 1, 2, 2, 3, 3, 4)$. Its median is $S_4 = 2$.

Sample Input 2

Copy

```
1
58
```

Sample Output 2

Copy

```
58
```

In this case, $S = (58)$.

2026-04-19 (Sun)
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